

Assigned in: Lesson 3

Due before: Lesson 4

For this lab we will create a hierarchy of several classes: IDevice (the parent), IPod, iPad, and iPhone.

The parent class – IDevice – has a concrete method called **getPurpose()** which returns its purpose (a String). The “purpose” instance variable is set in the constructor. The IDevice class also has an abstract method called **printDetails()** which prints all of the child class’s instance variables.

Child classes of IDevice:

- IPod: the purpose of this iDevice is “music”
- iPad: the purpose of this iDevice is “learning”
- iPhone: the purpose of this iDevice is “talking”

Note: the child classes also contain instance variables, constructor parameters, accessor methods, and mutator methods for several other data members:

- IPod: (int) number of songs stored, (double) maximum volume in decibels
- iPad: (boolean) has a case, (String) operating system version
- iPhone: (double) number of minutes remaining on phone plan, (String) carrier

Also, each of these four classes also overrides the **toString()** method to return all of the instance variables in a single String. Use the `@Override` annotation. Child classes’ **toString()** methods must also call their parent’s **toString()** method.

Furthermore, each of these four classes also overrides the **equals()** method. IPods with the same number of songs stored are considered equal; iPads with the same operating system version are considered equal; iPhones which have the same amount of minutes remaining as each other are considered equal. Use the `@Override` annotation when overriding **equals()**.

Remember to override the **hashCode()** method too for each class, whenever you override **equals()**.

Continuing from above, add more data and methods as described below.

Extend the iPhone class; it has a child called iPhone17.

The iPhone17 class also contains instance variables, constructor parameters, accessor methods, and mutator methods for several other data members:

- (boolean) high-resolution camera
- (int) gigabytes of memory

Also, the iPhone17 class overrides the **toString()** method to return all of the object data in a String. Use the `@Override` annotation. This **toString()** method must also call its parent’s **toString()** method.

Furthermore, this class also overrides **equals()** (and therefore also **hashCode()**). iPhone17 objects that have the same amount of minutes remaining on their phone plan are considered equal, but only if they also have the same value for “high-resolution camera”. Use the `@Override` annotation.

Remember to override **hashCode()** properly too for this class.

Create a Main class with the main() method. This method creates three objects of each of your four classes, and tells if your code looks correct or not (copy/paste this code BUT ALSO FIX ALL THE STYLE VIOLATIONS):

```
public class Main {
    public static void main(final String[] args) {
        // Create IPod objects
        final IPod ipod1;
        final IPod ipod2;
        final IPod ipod3;
        ipod1 = new IPod(300, 80.0); // 300 songs, max volume 80.0 dB
        ipod2 = new IPod(400, 85.0); // 400 songs, max volume 85.0 dB
        ipod3 = new IPod(300, 70.0); // 300 songs, max volume 70.0 dB

        // Test equality and inequality for IPod
        System.out.println("IPod Equality Test:");
        if (!ipod1.equals(ipod2)) {
            System.out.println("CORRECT: ipod1 is not equal to ipod2");
        } else {
            System.out.println("INCORRECT: ipod1 should not be equal to ipod2");
        }

        if (ipod1.equals(ipod3)) {
            System.out.println("CORRECT: ipod1 is equal to ipod3");
        } else {
            System.out.println("INCORRECT: ipod1 should be equal to ipod3");
        }
        System.out.println();

        // Create IPad objects
        final IPad ipad1;
        final IPad ipad2;
        final IPad ipad3;
        ipad1 = new IPad(true, "iPadOS 15"); // Has case, OS version iPadOS 15
        ipad2 = new IPad(false, "iPadOS 14"); // No case, OS version iPadOS 14
        ipad3 = new IPad(true, "iPadOS 15"); // Has case, OS version iPadOS 15

        // Test equality and inequality for IPad
        System.out.println("IPad Equality Test:");
        if (!ipad1.equals(ipad2)) {
            System.out.println("CORRECT: ipad1 is not equal to ipad2");
        } else {
            System.out.println("INCORRECT: ipad1 should not be equal to ipad2");
        }
    }
}
```

```
if (ipad1.equals(ipad3)) {
    System.out.println("CORRECT: ipad1 is equal to ipad3");
} else {
    System.out.println("INCORRECT: ipad1 should be equal to ipad3");
}
System.out.println();

// Create iPhone objects
final iPhone iphone1;
final iPhone iphone2;
final iPhone iphone3;
iphone1 = new iPhone(120.0, "Verizon"); // 120 minutes, carrier Verizon
iphone2 = new iPhone(180.0, "T-Mobile"); // 180 minutes, carrier T-Mobile
iphone3 = new iPhone(120.0, "AT&T"); // 120 minutes, carrier AT&T

// Test equality and inequality for iPhone
System.out.println("iPhone Equality Test:");
if (!iphone1.equals(iphone2)) {
    System.out.println("CORRECT: iphone1 is not equal to iphone2");
} else {
    System.out.println("INCORRECT: iphone1 should not be equal to iphone2");
}

if (iphone1.equals(iphone3)) {
    System.out.println("CORRECT: iphone1 is equal to iphone3");
} else {
    System.out.println("INCORRECT: iphone1 should be equal to iphone3");
}

System.out.println();

// Create iPhone17 objects
final iPhone17 iphone16_1;
final iPhone17 iphone16_2;
final iPhone17 iphone16_3;
iphone17_1 = new iPhone17(100.0, "Verizon", true, 512); // 100 minutes, high-res camera, 512 GB
iphone17_2 = new iPhone17(100.0, "Verizon", true, 256); // 100 minutes, high-res camera, 256 GB
iphone17_3 = new iPhone17(100.0, "Verizon", false, 512); // 100 minutes, no high-res camera, 512 GB

// Test equality and inequality for iPhone17
System.out.println("iPhone17 Equality Test:");
if (iphone17_1.equals(iphone17_2)) {
    System.out.println("CORRECT: iphone17_1 is equal to iphone17_2");
} else {
    System.out.println("INCORRECT: iphone17_1 should be equal to iphone17_2");
}

if (!iphone17_1.equals(iphone17_3))
```

```
        System.out.println("CORRECT: iphone17_1 is not equal to iphone17_3");  
    } else {  
        System.out.println("INCORRECT: iphone17_1 should not be equal to iphone17_3"); }  
    System.out.println();  
  
}  
}
```